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Utilization of waste fibers in stone matrix asphalt mixtures

Bradley J. Putman[a,*], Serji N. Amirkhanian[b,1]

[a] *Asphalt Rubber Technology Service, Department of Civil Engineering, Clemson University, 2002 Hugo Dr., Clemson, SC 29634, USA*
[b] *Department of Civil Engineering, Clemson University, 110 Lowry Hall, Clemson, SC 29634, USA*

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Abstract

Waste fibers produced from manufacturing processes such as scrap tire processing and automotive carpet manufacturing are sometimes used in other applications, but commonly are disposed of in landfills. If these fibers could be beneficially utilized in any application, it would reduce the load on the nation's landfills. Also, since these are waste materials, the cost of using these fibers compared to fibers manufactured for a specific application could be considerably less. The major objective of this research was to determine the feasibility of utilizing waste tire and carpet fibers in stone matrix asphalt (SMA). Many states utilize such rut resistant SMA mixtures on heavily traveled highways. Fibers are included in SMA mixtures as a stabilizing additive to prevent excessive draindown caused by relatively high contents of polymer modified asphalt binder. The common types of fiber used in SMA include cellulose and mineral fibers. This study compared the performance of SMA mixtures containing waste tire and carpet fibers with mixes made with commonly used cellulose and other polyester fibers produced specifically for use in hot mix asphalt (HMA). No significant difference in permanent deformation or moisture susceptibility was found in mixtures containing waste fibers compared to cellulose or polyester. Also, the tire, carpet, and polyester fibers significantly improved the toughness of the mixtures compared to the cellulose fibers.

Keywords: Asphalt; Stone matrix asphalt; Stone mastic asphalt; Fibers; Scrap tires; Waste materials; Carpet waste

* Corresponding author. Tel.: +1-864-656-6185; fax: +1-864-656-6186.
E-mail address: bputman@clemson.edu (B.J. Putman).
[1] Tel.: +1-864-656-3316; fax: +1-864-656-2670.

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1. Introduction

In some parts of the United States, the disposal of waste fibers produced from numerous manufacturing processes is an economical and environmental problem for companies and municipalities (e.g., cost, landfill space, etc.). These fibers are sometimes used in other applications, but commonly they are disposed of in landfills. Two industries that produce such waste fibers are the tire processing industry and the automotive carpet industry. If these fibers could be beneficially utilized in any application, it would reduce the burden on diminishing landfill space. Also, using such waste materials could be economically beneficial compared to fibers manufactured for a specific application.

Processing scrap tires involves the shredding and/or grinding of the tires into smaller particles called crumb rubber. A typical tire contains approximately 60% rubber, 20% steel, and 20% fiber. The crumb rubber is used for many applications ranging from rubberized asphalt to playground construction. The steel is typically sold to steel manufacturers where it is recycled. However, in most cases the polyester fibers have no specific end use and are disposed of in landfills. For instance, three tractor trailer loads of scrap tire fiber are hauled to the landfill every 2 weeks from one company's scrap tire processing plant. With approximately 33,000,000 tires processed into crumb rubber each year, that makes for an abundance of fiber that is going to waste (RMA, 2002).

In the production of automotive carpets and fabrics such as headliners, seat backs, and door panels there is a waste stream resulting from the process of shearing the woven fabrics to size. This shearing process creates waste in the form of fine polyester particles that are collected and form tufts of lint. A small portion of this lint is recycled, but yet most of this waste is disposed of elsewhere.

Fibers have been used in asphalt mixtures for two main reasons: (1) to increase the toughness and fracture resistance of hot mix asphalt (HMA); and (2) to act as a stabilizer to prevent draindown of the asphalt binder (Hansen et al., 2000). There has been limited use of fibers to increase toughness, even though research has shown that the addition of polyester fibers does increase the toughness of asphalt mixtures (Freeman et al., 1989). Fibers, such as cellulose and mineral fibers, have been used in large part as a stabilizer in such asphalt mixtures as stone matrix asphalt (SMA) and open graded friction courses (OGFC).

Stone matrix asphalt was developed in Europe in the 1970s as a rut resistant asphalt mixture. This type of rut resistant mixture has been used by several states in the US since 1991 and its popularity is growing (Brown et al., 1997). SMA is defined as a HMA prepared with a gap-graded aggregate gradation in order to maximize the asphalt binder content and coarse aggregate fraction (Brown and Cooley., 1999). The large coarse aggregate fraction provides rut resistance through stone-to-stone contact, while the high binder content (typically 5.5–7.0% by weight of mixture) provides durability through an increased film thickness around the aggregate particles.

In order to obtain such a high binder content without excessive draindown, mineral filler and stabilizers are added to the mixture. Examples of mineral fillers include fly ash, hydraulic cement, hydrated lime, and rock dust, which all have a high percentage of particles passing the 75 μm sieve (55–100%). Polymer modified asphalt binders are typically used in the production of SMA mixtures due to their rut resistant properties. The increased viscosity of these binders is also beneficial to minimize draindown in the mix. To further aid in

Table 1
Aggregate gradation and combined gradation with SC DOT specifications for SMA

Size	Passing (% by weight)							
	6 M stone	67 M stone	789 M stone	Regular screenings	Mineral filler	Hydrated lime	Combined gradation	Specifications
25.4 mm	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100
19.0 mm	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100
12.5 mm	42.5	53.1	99.8	100.0	100.0	100.0	87.1	85–90
9.5 mm	9.6	23.4	82.5	100.0	100.0	100.0	69.5	60–80
4.75 mm	0.6	1.5	23.5	96.7	100.0	100.0	31.4	25–32
2.36 mm	0.4	0.8	4.4	87.1	100.0	100.0	19.6	18–24
600 μm	0.3	0.7	1.4	49.7	100.0	100.0	14.9	12–20
300 μm	0.3	0.6	0.8	23.7	98.0	100.0	12.3	9–15
75 μm	0.25	0.44	0.59	15.56	88.00	80.00	10.4	8–12
Percent in mixture	10	15	57	8	9	1		

the reduction of draindown, fibers are typically included in SMA mixtures as a stabilizing additive. The most common types of fibers used in SMA include cellulose and mineral fibers. Some of these fibers, depending on the source, could be relatively expensive.

2. Materials, methods, and experimental design

The SMA mixtures tested in this study were made with a PG 76–22 asphalt, mineral aggregate, and stabilizing fibers. Crushed granite aggregate was blended to meet the South Carolina Department of Transportation's (SC DOT) 12.5 mm SMA gradation (Table 1 and Fig. 1). To meet the required gradation, the 19.0 mm stone was removed from the 6 M and

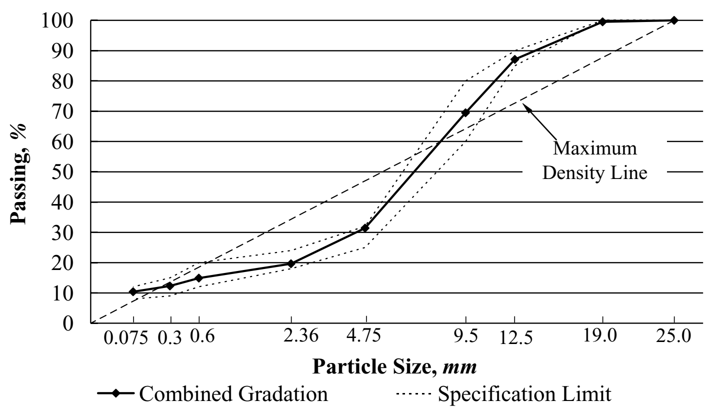

Fig. 1. Aggregate gradation with specification limits and maximum density line (sieve sizes raised to the 0.45 power).

Table 2
Particle size distribution of cellulose fibers

Size (μm)	Passing (%)
106	20–40
425	55–75
850	75–95

Determined by: Alpine Air Draught Sieve.

67 stone. Mineral filler was also added to the mix to meet the passing 75 μm requirement. Hydrated lime was included at 1.0% by weight of the dry aggregate as an anti-strip additive.

The fibers that were included in the study included cellulose, polyester, scrap tire, and waste carpet fibers. The particle size distribution of the cellulose fibers is included in Table 2. The polyester fibers included in the study were 4.5 denier monofilament fibers obtained from recycled raw materials that were cut to 6.35 mm in length. The tire fibers were collected from the cryogenic processing of scrap tires. The fibers ranged from 3 to 13 mm in length, contained trace amounts of rubber dust and were mostly free of steel. The carpet fibers were in the form of fiber tufts similar to mineral fibers that needed to be torn into smaller tufts.

The aggregate and asphalt binder sources remained constant for this study. The fiber content was also held constant at 0.3% by weight of total mixture. This fiber percentage was selected after reviewing the literature of SMA practices. The fiber type was varied yielding four different mix designs followed by draindown, moisture susceptibility, and permanent deformation testing of each mixture (Fig. 2). For this study, the mixture containing cellulose fibers was used as the control mix due to the wide use of cellulose fibers in SMA mixes.

Mix designs were completed following Superpave guidelines and by varying the binder content from 5.5 to 7.0% by weight of mixture. Each mix design specimen was compacted with 50 gyrations of the Superpave Gyratory Compactor per SC DOT procedures. Bulk specific gravity (BSG) was determined using the CoreLok method for each specimen (Table 3).

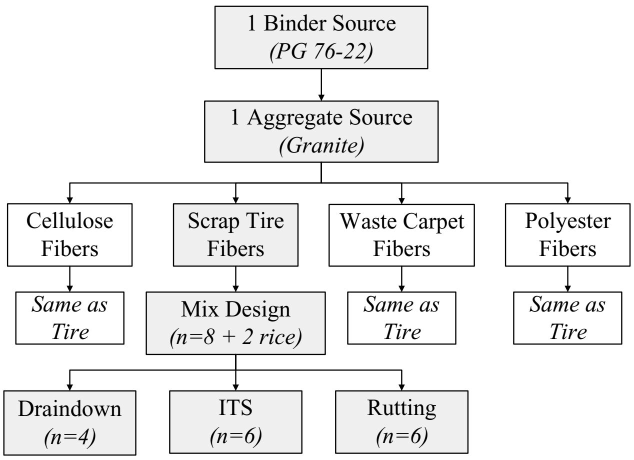

Fig. 2. Experimental design.

Table 3
Properties of test mixtures

Fiber	BSG[a]	VMA[b] (%)	VFA[c] (%)	OAC[d] (%)
Cellulose	2.297	17.2	76.7	5.95
Tire	2.301	17.0	76.4	5.85
Carpet	2.295	16.2	75.3	5.50
Polyester	2.296	16.8	76.2	5.80

[a] Bulk specific gravity.
[b] Voids in mineral aggregate.
[c] Voids filled with asphalt.
[d] Optimum asphalt content.

The CoreLok was used due to its applicability to coarse mixtures such as SMA (Cooley et al., 2003). The optimum asphalt content (OAC) of each mixture was determined at 4.0% air voids (Table 3).

After determining the OAC of each mixture, draindown tests were performed per AASHTO T 305 *determination of draindown characteristics in uncompacted asphalt mixtures*. Draindown was tested by placing the uncompacted mixture in a basket in an oven at the mixing temperature of the binder (162 °C) and at 177 °C per AASHTO T 305. The draindown was calculated as the percentage of binder that drained out of the basket compared to the original weight of the sample.

Draindown was also tested at binder contents exceeding the OAC to determine the stabilizing capacity of each fiber. Most states require that the draindown of SMA mixtures not exceed 0.3% by weight of the mixture. The binder contents used in this portion of the study started at 6.5% (by weight of mixture) and increased by 1.0% until a draindown of 0.3% was reached. The stabilizing capacity of each fiber was determined as the binder content at which the draindown reached 0.3%.

Moisture susceptibility was conducted by comparing the indirect tensile strength (ITS) of three 100 mm diameter × 63.5 mm tall specimens conditioned in 60 ± 1 °C water for 24 h to the ITS of three specimens, of the same dimensions, dry conditioned at 25 ± 1 °C (modified ASTM D 4867). The ITS specimens were compacted to 6–8% air voids with a Marshall hammer. Each specimen was loaded to failure and the following parameters were evaluated:

1. Indirect tensile strength (ITS):

$$\mathrm{ITS} = \frac{2P_{\max}}{\pi td}, \tag{1}$$

where $P_{\max}$ is peak load (N), t the average height of specimen (mm) and d the diameter of specimen (mm).

2. Tensile strength ratio (TSR):

$$\mathrm{TSR}\ (\%) = \frac{\mathrm{ITS_{wet}}}{\mathrm{ITS_{dry}}} \times 100. \tag{2}$$

3. Toughness: toughness was defined as the area under the tensile stress–deformation curve up to a deformation of twice that incurred at maximum tensile stress (Fig. 3) (Freeman et al., 1989).

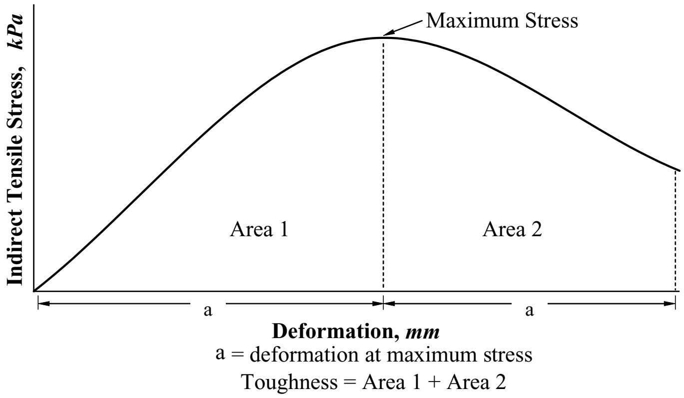

Fig. 3. Definition of toughness.

4. Percentage of toughness loss (PTL):

$$\text{PTL}\ (\%) = \frac{\text{toughness}_{\text{dry}} - \text{toughness}_{\text{wet}}}{\text{toughness}_{\text{dry}}} \times 100. \tag{3}$$

Finally, the rutting resistance of each mixture was determined using the Asphalt Pavement Analyzer and SC DOT procedures by testing six 150 mm diameter × 75 mm tall cylinders with air voids ranging from 3 to 5%. The specimens were conditioned at 76 °C for 4 h prior to testing. Upon completion of the test, the dynamic stability of each mixture was calculated by determining the slope of the rutting curve after 1000 cycles (Fig. 4). The slope was then used to calculate the dynamic stability as the number of cycles required to cause 1 mm of deformation.

3. Results and discussion

3.1. Statistical considerations

Results of the indirect tensile and rutting tests were statistically analyzed with a 5% level of significance. For these comparisons, it should be noted that all specimens were produced at optimum asphalt content (OAC). The effects on the physical properties of the mixtures (i.e., rutting, indirect tensile strength, and toughness) caused by the fibers include the effects of the different OACs.

3.2. Draindown

At the OAC, there was no statistical difference in the draindown of any of the mixtures. Each mix had a draindown value of 0% at both test temperatures (162 and 177 °C). This indicates that in all mixtures, at the OAC, each fiber is performing its function as a stabilizing additive.

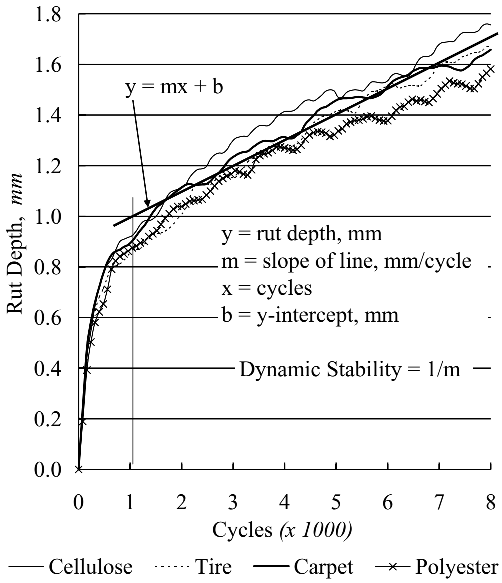

Fig. 4. Average rut depth development and definition of dynamic stability.

There were, however, some statistical differences in the performance of each fiber at binder contents greater than the OAC (Fig. 5). It is evident that all of the fibers provide significant stabilization of the mixture as compared to the mixture containing no fibers. The mixture without fibers reached 0.3% draindown at 6.53% binder (by weight of mix). All of the fibers exceeded 10% binder before reaching 0.3% draindown. The cellulose fiber had the highest stabilizing capacity at 13.36%, followed by the polyester fiber (10.86%), the carpet fiber (10.34%), and the tire fiber reached the draindown limit at 10.11% (Fig. 5).

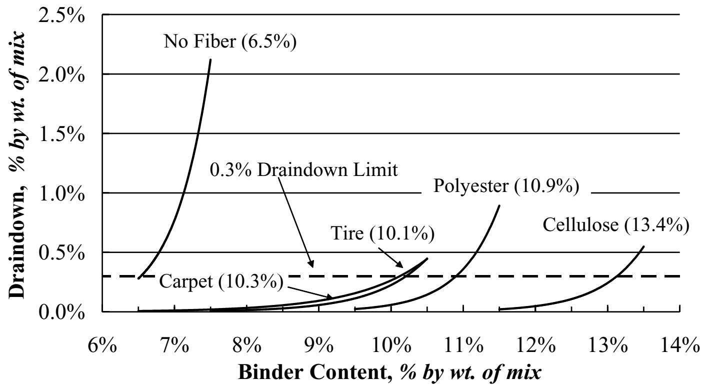

Fig. 5. Draindown results at varying binder contents.

Table 4
Stabilizing capacities of fibers

Fiber	Stabilizing capacity (%)
Cellulose	13.4
Tire	10.1
Carpet	10.3
Polyester	10.9
No fiber	6.5

These results indicate that the waste tire and carpet fibers have similar stabilizing capacities as compared with the polyester fibers. The cellulose fibers have a much higher stabilizing effect, which can be attributed to the absorptive nature of the cellulose compared to the synthetic polyester fibers.

3.3. Moisture susceptibility

There was no significant difference in the average dry indirect tensile strength (ITS) values between the different types of fibers (Table 5). The mix containing the cellulose fibers produced the highest dry ITS followed by the polyester fibers, the carpet fibers, and finally, the tire fibers produced the lowest dry ITS.

The average wet ITS values were not significantly different for any of the mixtures (Table 4). These results greatly exceeded the minimum requirement of 448 kPa set by the SC DOT.

The tensile strength ratio (TSR) for all of the mixtures exceeded the minimum value of 85% required by the SC DOT (Table 5). This indicates that these fibers do not cause a decrease in the strength of the SMA due to the intrusion of water into the mix. The mixtures containing the waste fibers (carpet and tire) had a TSR above 100%, followed by the polyester and cellulose fibers.

Analysis of the toughness results of both the dry and the wet specimens indicated that the mixture containing the cellulose fibers had significantly lower toughness values than the tire, carpet, and polyester fibers (Table 6). This shows that the polyester, tire and carpet fibers have a better ability to bridge cracks that have developed within the mix during loading; therefore, increasing the mixture toughness. This could also indicate that there is a stronger bond between the polyester fibers and the asphalt binder than there is with the cellulose fibers.

Table 5
Average indirect tensile strength and tensile strength ratio results

Fiber	Dry ITS[a] (kPa)	Wet ITS[a] (kPa)	TSR[b] (%)
Cellulose	873	821	94.1
Tire	838	846	100.9
Carpet	856	872	101.8
Polyester	865	863	99.8

[a] Indirect tensile strength.
[b] Tensile strength ratio.

Table 6
Average toughness results

Fiber	Dry toughness (N/mm)	Wet toughness (N/mm)	PTL[a] (%)
Cellulose	3.9	3.4	12.7
Tire	4.0	4.0	−0.2
Carpet	4.5	3.9	14.7
Polyester	4.5	4.2	7.3

[a] Percentage toughness loss.

Table 7
Average permanent deformation results

Fiber	Rut depth (mm)	Dynamic stability (cycles/mm)
Cellulose	1.76	9,637
Tire	1.67	8,925
Carpet	1.66	10,987
Polyester	1.60	10,299

The percentage of toughness loss results indicate that the mixture containing the tire fibers did not lose toughness when conditioned in water (Table 6). The carpet fiber mix had the highest PTL (14.7%), followed by the cellulose and polyester fibers.

3.4. Permanent deformation

Analysis of the rut depth indicates that there is not a significant difference in the rutting resistance between any of the fibers (Table 7). The mixture containing the polyester fiber resulted in the smallest rut depth followed by the carpet, tire, and cellulose fibers, respectively. Fig. 4 shows the average rut depth development of the different mixtures.

The dynamic stability results revealed that there was no significant difference between any of the mixtures (Table 7). It should be noted that these dynamic stability values are only applicable to the APA with a wheel load of 445 N and a hose pressure of 689 kPa.

4. Conclusions

Based on this limited study of the utilization of waste tire and carpet fibers in stone matrix asphalt (SMA) mixtures, the following findings were made:

1. The tire and carpet fibers, obtained from waste streams, were effective in preventing excessive draindown of the SMA mixtures.
2. The mixtures containing the waste tire and carpet fibers greatly exceeded the indirect tensile strength requirements set forth by the SC DOT and many other states in the US.
3. The tensile strength ratios for the mixes containing the tire and carpet fibers were greater than 100% (100.9 and 101.8%, respectively). This indicates that these fibers do not cause the mixture to weaken when exposed to moisture.

4. The addition of tire and carpet fibers increases the toughness of the SMA mixtures as do polyester fibers when compared to the cellulose fibers.
5. The mix containing tire fibers did not lose any toughness when conditioned in water.
6. There was no significant difference in permanent deformation (rut depth or dynamic stability) of the waste fibers as compared with the other fibers included in the study.
7. The mixtures containing tire, carpet, and polyester fibers had lower optimum asphalt contents than the mix containing cellulose fibers (0.1, 0.45, and 0.15% lower, respectively). By reducing the amount of the most expensive constituent of an asphalt mixture (asphalt binder), the result could be more cost effective pavements.

These findings lead to the conclusion that waste tire and carpet fibers can be viable options for use as stabilizing additives in stone matrix asphalt mixtures. In addition to preventing draindown, they also increase the toughness of the mixture while maintaining the moisture susceptibility performance and permanent deformation performance compared to SMA mixtures made with fibers commonly used in industry. Utilizing these waste fibers in SMA mixtures would potentially be more cost effective and valuable landfill space would be spared at the same time.

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